SECURE FILE SHARING OVER WLAN

Project Abstract

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ABSTRACT

Secure File Sharing over WLAN is a software application designed to enable secure and efficient file transfer between devices connected to the same Wireless Local Area Network (WLAN). The system eliminates the need for external internet-based file-sharing services by allowing devices on the same network to communicate through a local client-server architecture. Users can authenticate before initiating a transfer, select files for sharing, and transmit them securely over the WLAN.

To enhance security, the proposed system employs AES-256-GCM encryption to protect file contents during transmission. At the receiving end, the encrypted data is decrypted using the appropriate key. SHA-256 hashing can additionally be used to verify file integrity by comparing the hash values of the original and received files. The application provides a simple interface for connecting to the receiver, selecting files, monitoring transfers, and confirming successful delivery. The project demonstrates the practical application of WLAN communication, client-server networking, authentication, encryption, and data integrity in mobile computing.

Keywords: WLAN, Secure File Transfer, Client-Server Architecture, Authentication, AES-256-GCM, SHA-256, Data Integrity, Mobile Computing